TERM 2

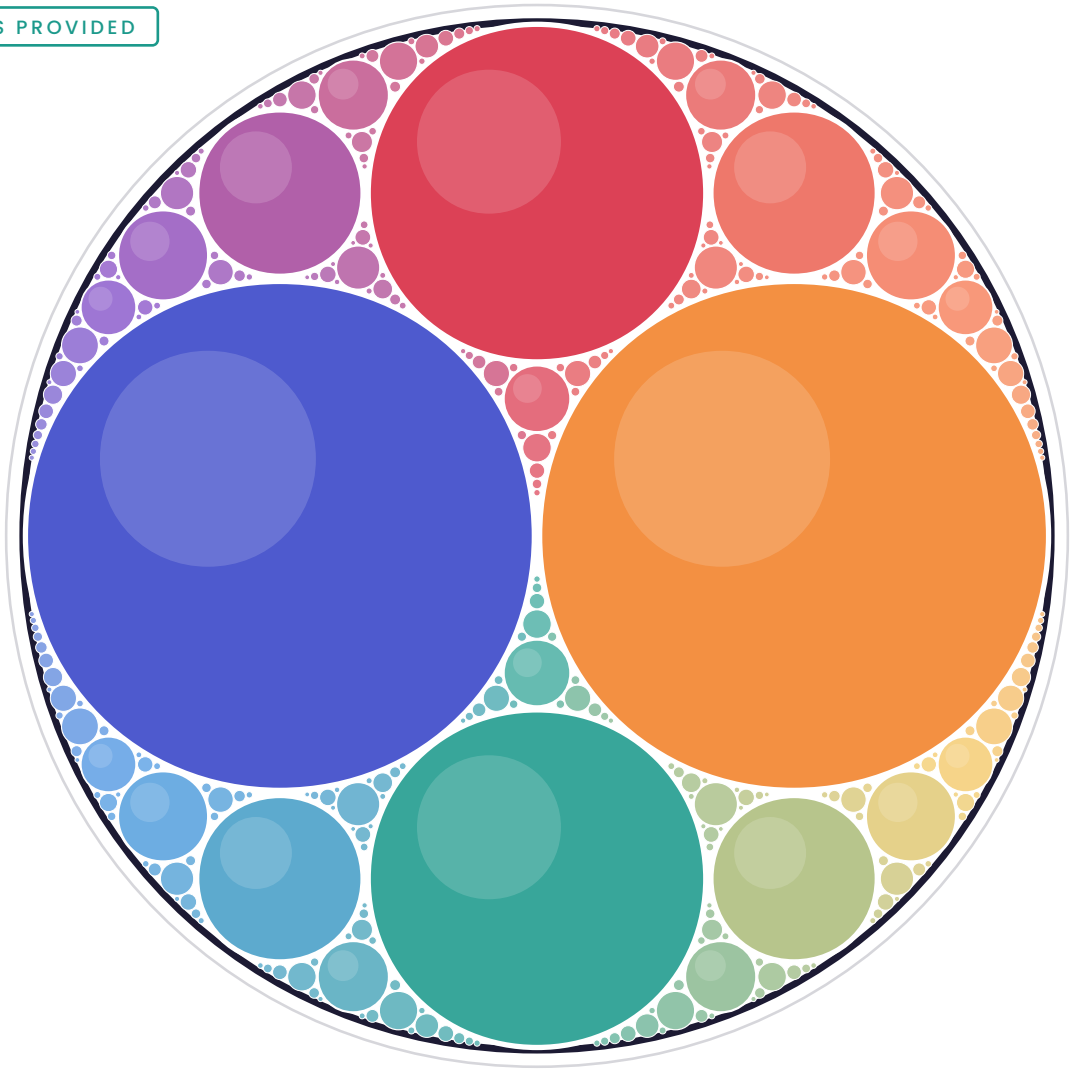
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 8

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

APOLLONIAN GASKET
every circle touches three others; every curvature is a whole number

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

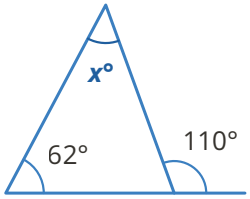
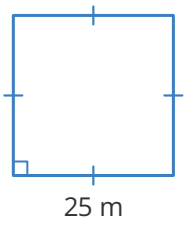
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the perimeter of this square.  	FORMULA $P = 4s$ COPY SUBSTITUTE ANSWER
3 Evaluate. $24 \div (-6) + 7$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Solve. $5k = 45$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Find the probability that the student does not pass. $P(\text{pass}) = \frac{4}{5}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

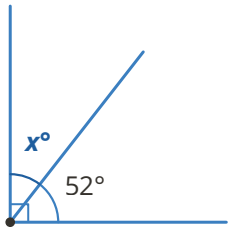
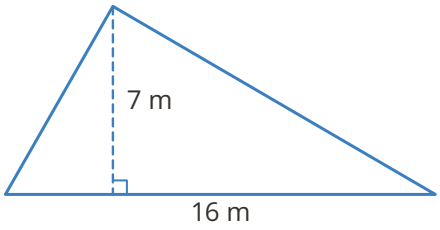
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 60% of 35 kg	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Write 3:40 pm in 24-hour time.	FORMULA 24-hour time: pm hours add 12 (1:00 pm = 13:00); 12:00 am = 00:00 COPY SUBSTITUTE ANSWER
5 Find the median of these scores. Scores: 7, 3, 9, 5, 8	FORMULA Median = middle score when the scores are in order COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

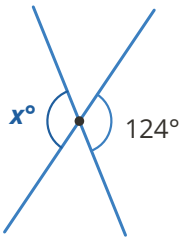
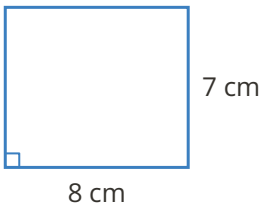
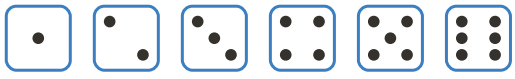
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Vertically opposite angles are equal COPY SUBSTITUTE ANSWER
2 Find the area of this rectangle.  	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER
3 A netball shooter scored 21 goals from 25 shots. What percentage of the shots were goals?	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Expand. $3(2x + 4)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling an even number. 	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

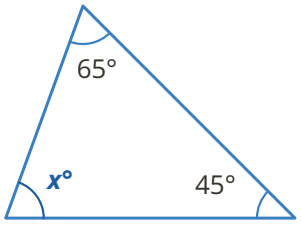
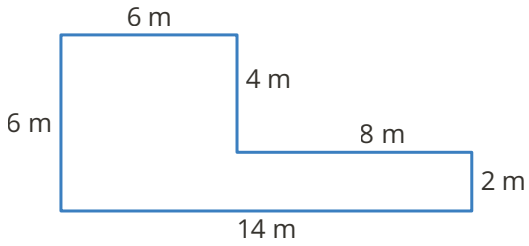
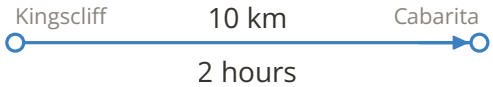
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this shape.  	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{1}{6}$ of 54	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 A walker walks from Kingscliff to Cabarita. Find the average speed.  	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Complete the table for $y = 4x$. x | 1 | 2 | 3 | 4 y | 4 | | | 16	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

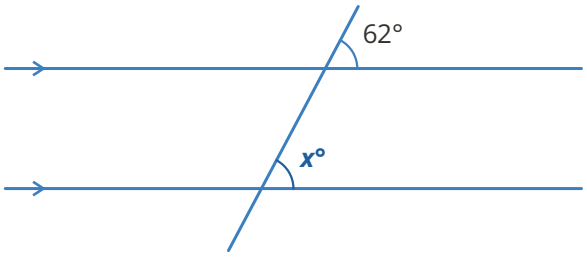
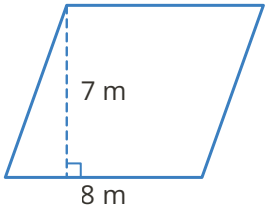
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Corresponding angles are equal COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram.  	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Write as a fraction in simplest form. 0.125	FORMULA Decimal to fraction: write over 10, 100 or 1000, then simplify COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $y = 9$. $5(y - 2)$	FORMULA $5(y - 2)$ COPY SUBSTITUTE ANSWER
5 Find the mean of these marks. Marks: 45, 52, 48, 50, 55	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

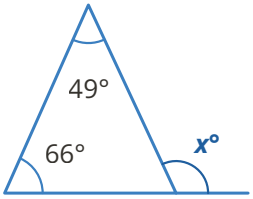
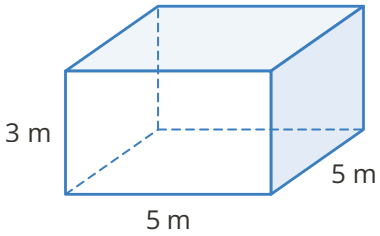
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Calculate. 0.038×100	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Simplify. $24 : 16$	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 Find the range of these scores. Scores: 40, 38, 45, 33, 41	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

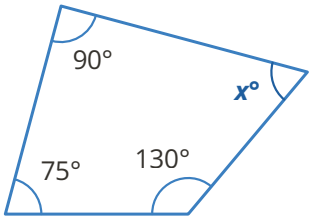
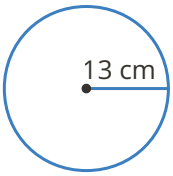
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = 2\pi r$ COPY SUBSTITUTE ANSWER
3 Evaluate. 7^2	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Solve. $\frac{x}{3} + 2 = 7$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 A spinner was spun 50 times. Find the experimental probability of red. Colour: Red, Blue, Green Frequency: 14, 26, 10	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

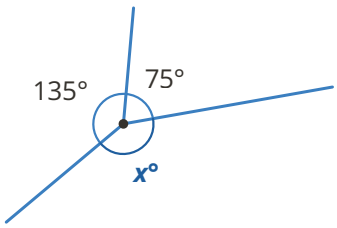
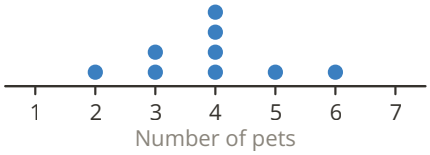
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the perimeter of this rectangle. 	FORMULA $P = 2(l + w)$ COPY SUBSTITUTE ANSWER
3 Calculate. 40% of \$75	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 3 kg of potatoes cost \$12. Find the price per kilogram.	FORMULA Unit rate = amount ÷ number of units COPY SUBSTITUTE ANSWER
5 The dot plot shows the number of pets students in a class own. Find the mode. 	FORMULA Mode = the most common score COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

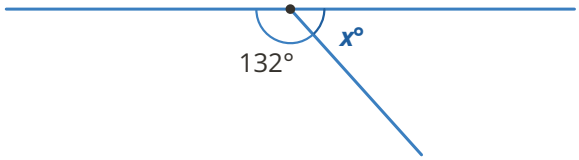
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Convert 2.5 m^2 to square centimetres.	FORMULA $1\ m^2 = 10\ 000\ cm^2$ COPY SUBSTITUTE ANSWER
3 24 of the 30 students in a class walk or ride to school. What percentage is that?	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Expand. $2(5a - 4)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the probability that the target is not hit. $P(\text{hit}) = \frac{7}{9}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

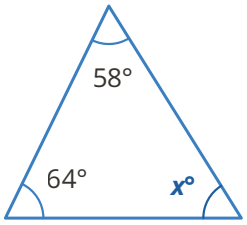
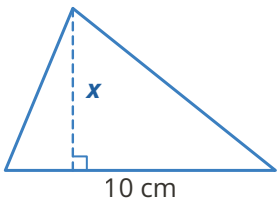
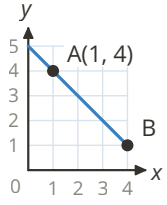
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 The area of this triangle is 30 cm². Find x. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{3}{8}$ of 96 m	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Share \$200 in the ratio 1 : 4. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = 5 - x$. Write the coordinates of B. 	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

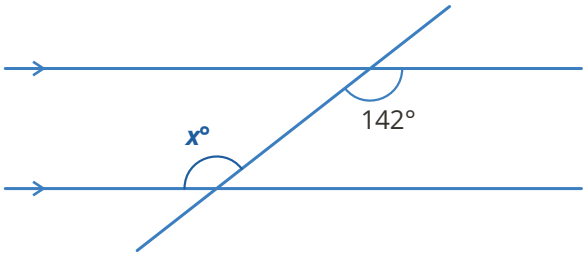
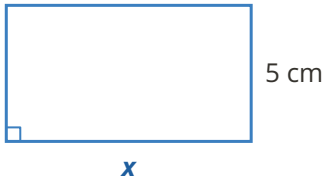
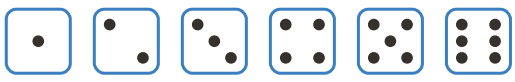
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Alternate angles are equal COPY SUBSTITUTE ANSWER
2 The area of this rectangle is 45 cm². Find x. 	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER
3 Evaluate. $18 - 6 \times 2$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $n = 4$. $\frac{12}{n} + 3$	FORMULA $\frac{12}{n} + 3$ COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a number greater than 4. 	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

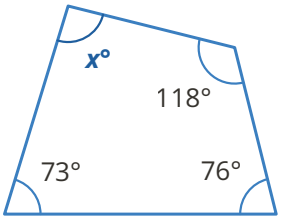
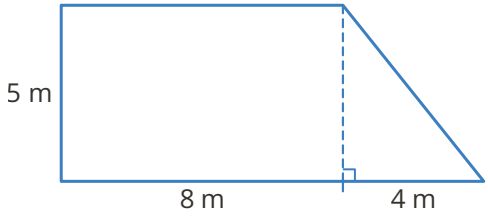
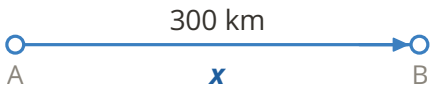
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this shape. 	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Write as a percentage. 0.72	FORMULA Decimal to percentage: decimal × 100% COPY SUBSTITUTE ANSWER
4 A train travels from A to B at an average speed of 60 km/h. Find x. 	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the mean of these temperatures. °C: 24, 31, 27, 30, 28	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

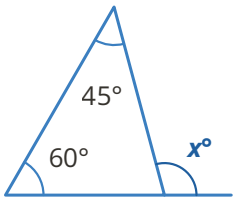
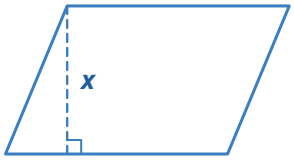
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 The area of this parallelogram is 54 cm². Find x. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 20% of 45 kg	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Solve. $\frac{x}{4} = 6$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Find the range of these marks. Marks: 61, 47, 55, 70, 52	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

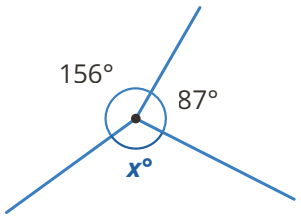
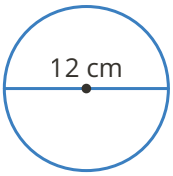
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER
3 Evaluate. $\sqrt[3]{1000} + \sqrt{36}$	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 A stall at the Kingscliff markets is open from 07:15 to 12:00. How long is it open?	FORMULA Time taken = finish time − start time (60 minutes = 1 hour) COPY SUBSTITUTE ANSWER
5 Heads came up 27 times in 90 tosses of a coin. Find the experimental probability of heads.	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

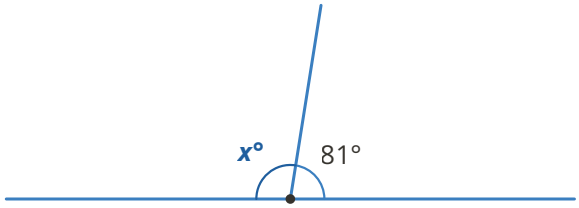
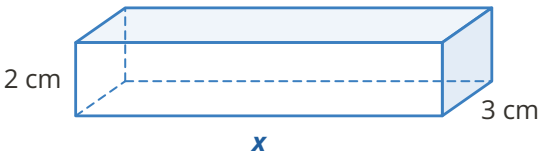
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 The volume of this rectangular prism is 60 cm^3. Find x. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. 42 out of 70	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Expand. $6(2 - y)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the median of these scores. Scores: 4, 9, 7, 3, 8, 5	FORMULA Median = middle score when the scores are in order COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

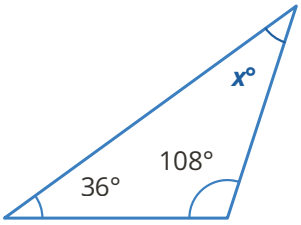
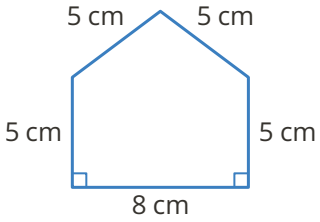
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the perimeter of this pentagon.  	FORMULA Perimeter = add the lengths of all the sides COPY SUBSTITUTE ANSWER
3 Calculate. $7.5 \div 10$	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Simplify. $6 : 9 : 15$	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 Find the probability that a goal is not scored. $P(\text{goal}) = \frac{3}{7}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

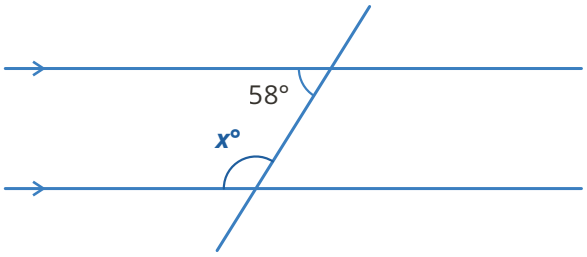
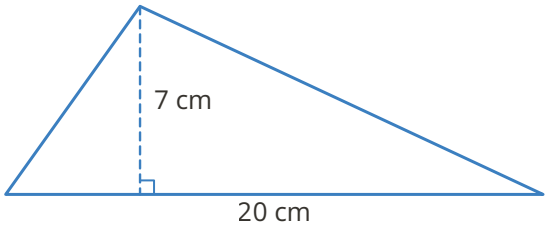
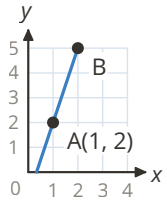
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Co-interior angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 A netball team won $\frac{5}{6}$ of its 18 games. How many games did it win?	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $a = 6$. $a^2 + 5$	FORMULA $a^2 + 5$ COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = 3x - 1$. Write the coordinates of B. 	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

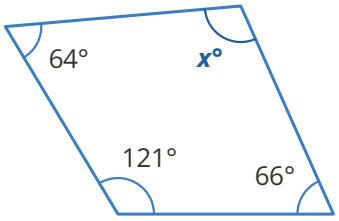
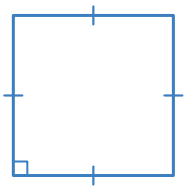
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this square.  	FORMULA $A = s^2$ COPY SUBSTITUTE ANSWER
3 Write as a fraction in simplest form. 80%	FORMULA Percentage to fraction: write over 100, then simplify COPY SUBSTITUTE ANSWER
4 Which bag of potatoes is the better buy? Bag A: 3 kg for $12 Bag B: 5 kg for $15	FORMULA Price per unit = price ÷ number of units; best buy = lowest price per unit COPY SUBSTITUTE ANSWER
5 Find the mean of these marks. Marks: 14, 11, 17, 12, 16	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

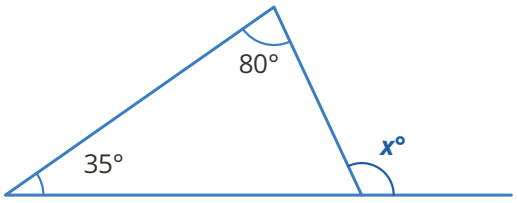
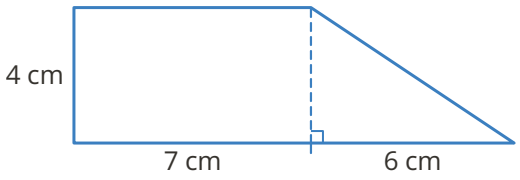
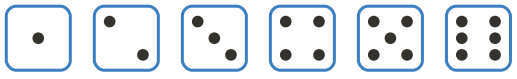
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the area of this shape. 	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. 30% of 90 L	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Solve. $7 + 3p = 25$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a multiple of 3. 	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

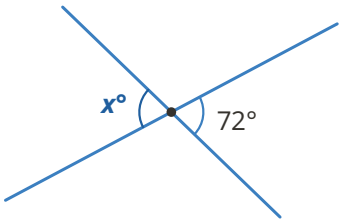
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Vertically opposite angles are equal COPY SUBSTITUTE ANSWER
2 Convert 600 cm to metres.	FORMULA 1 cm = 10 mm, 1 m = 100 cm, 1 km = 1000 m COPY SUBSTITUTE ANSWER
3 Evaluate. $36 \div (4 + 5) - 10$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Share \$360 in the ratio 7 : 5. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the range of these heights. cm: 104, 96, 118, 101, 99	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

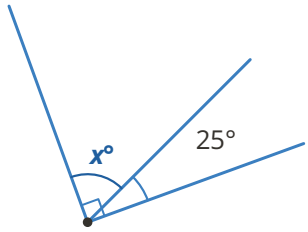
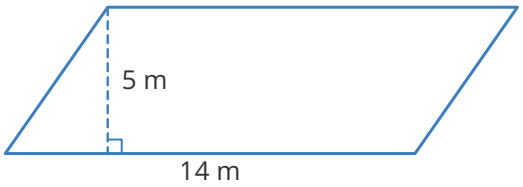
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Mia scored 18 out of 24 on a quiz. Write Mia's score as a percentage.	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Expand. $3(4p + 1)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the mode of these scores. 2 4 4 1 3 2 4 5	FORMULA Mode = the most common score COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

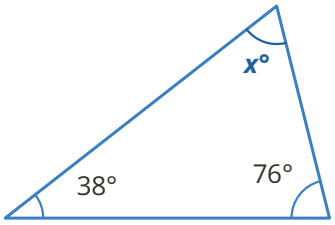
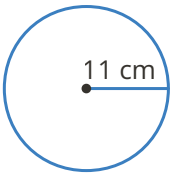
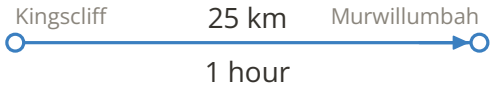
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = 2\pi r$ COPY SUBSTITUTE ANSWER
3 Evaluate. $\sqrt{81}$	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 A cyclist rides from Kingscliff to Murwillumbah. Find the average speed. 	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 A die was rolled 60 times. Find the experimental probability of rolling a 6. Number: 1, 2, 3, 4, 5, 6 Frequency: 8, 11, 9, 12, 10, 10	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

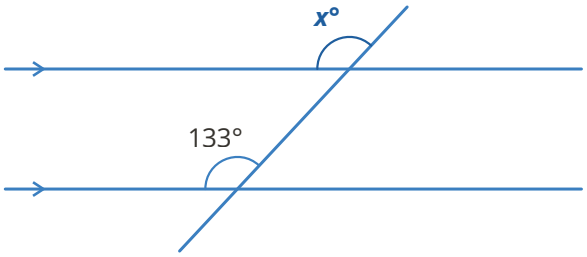
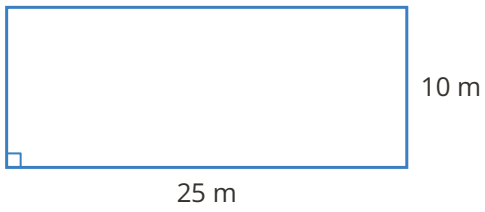
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Corresponding angles are equal COPY SUBSTITUTE ANSWER
2 Find the perimeter of this rectangle.  	FORMULA $P = 2(l + w)$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{5}{8}$ of \$120	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $k = 4$. $20 - 3k$	FORMULA $20 - 3k$ COPY SUBSTITUTE ANSWER
5 Find the probability that the marble is not blue. $P(\text{blue}) = \frac{5}{12}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

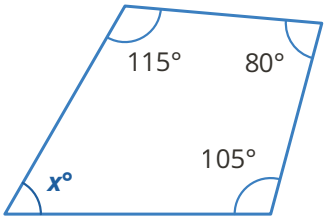
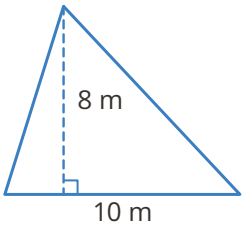
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Write as a decimal. 7%	FORMULA Percentage to decimal: percentage ÷ 100 COPY SUBSTITUTE ANSWER
4 Write 12:45 pm in 24-hour time.	FORMULA 24-hour time: pm hours add 12 (1:00 pm = 13:00); 12:00 am = 00:00 COPY SUBSTITUTE ANSWER
5 Complete the table for $y = 2x + 3$. x: 0, 1, 2, 3 y: 3, , , 9	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

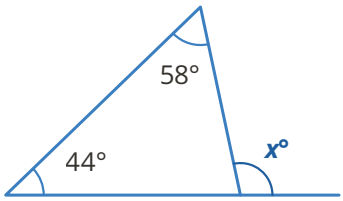
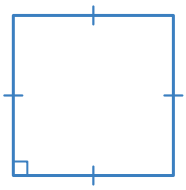
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the area of this square.  	FORMULA $A = s^2$ COPY SUBSTITUTE ANSWER
3 Calculate. 90% of 50 kg	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Simplify. 40 : 30	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 Find the mean of these goal tallies. Goals: 3, 5, 2, 6, 4, 4	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

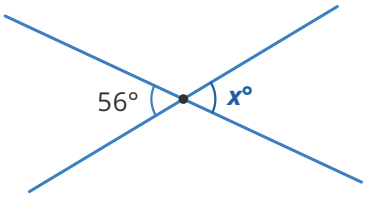
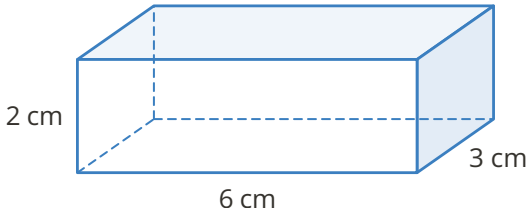
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Vertically opposite angles are equal COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism.  	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. 33 out of 60	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Solve. $2x + 9 = 25$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is yellow. Colour: Blue 3, Yellow 5, Red 4	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

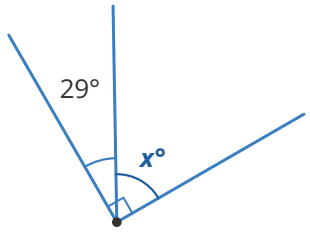
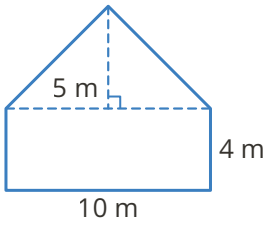
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 Find the area of this shape. 	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. 5.19×1000	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Share \$60 in the ratio 1 : 2. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 The dot plot shows the hours of sport some students played last week. Find the median. 	FORMULA Median = middle score when the scores are in order COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

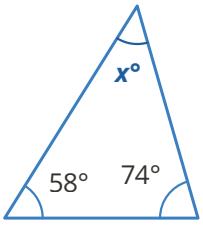
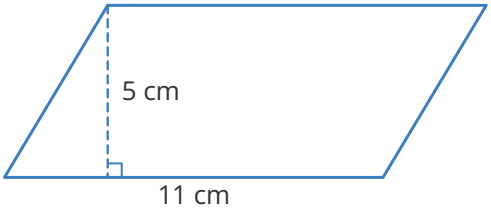
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{3}{4}$ of 60	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Which bottle of juice is the better buy? Bottle A: 2 L for $8 Bottle B: 3 L for $9	FORMULA Price per unit = price ÷ number of units; best buy = lowest price per unit COPY SUBSTITUTE ANSWER
5 Find the range of these maximum temperatures. °C: 19, 25, 22, 31, 27	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

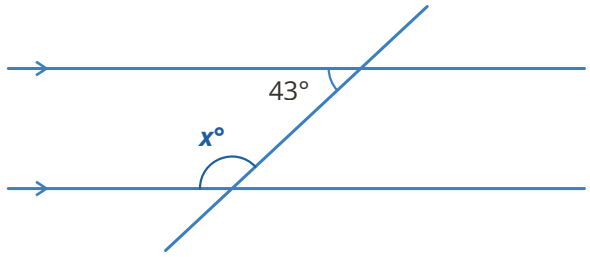
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Co-interior angles add to 180° COPY SUBSTITUTE ANSWER
2 Convert 4000 mL to litres.	FORMULA $1\ \text{cm}^3 = 1\ \text{mL},\ 1\ \text{L} = 1000\ \text{mL}$ COPY SUBSTITUTE ANSWER
3 Evaluate. 2^5	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Expand. $8(m + n)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 A netball goal shooter scored 32 goals from 40 shots. Find the experimental probability of scoring a goal.	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

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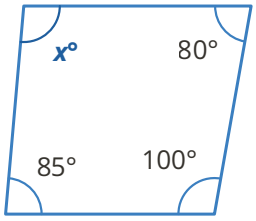
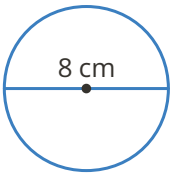
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER
3 Write as a decimal. $\frac{9}{25}$	FORMULA Fraction to decimal: numerator ÷ denominator COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $c = 3$ and $d = -5$. $7c - d$	FORMULA $7c - d$ COPY SUBSTITUTE ANSWER
5 Complete the table for $y = 3x + 2$. x: 0, 1, 2, 3 y: 2, 5, ___, ___	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER